Supplementary Material for "KESTREL: A Self-Checking, Retunable Edge Learner for Trustworthy Vibration Fault Diagnosis on Microcontrollers"

Bracketed numbers refer to the reference list of the paper; [8] is the earlier conference pipeline. All values are produced by the scripts at https://github.com/deth102/KESTREL (experiments e01–e27); the on-chip results come from the node firmware and the host scripts of experiments e19 and e20. CWRU results use the bearing orders ($\mathcal{F}_{22}$, Note S6) unless stated otherwise.

S1. The Configuration of [8] Under the Present Code

Table S1 evaluates the configuration of the earlier pipeline [8] and its variants with the present code on the x channel of bearing housing A: a 10-tree RF on KAIST with five classes (normal, unbalance, misalignment, inner race, outer race), both fault severities merged, a random stratified 70/15/15 split, and variants of the load subset, the window length and overlap, the feature set (the ten spectral features S10 of [8], or $\mathcal{F}_{16}$), and the forest implementation (scikit-learn or the original pure-Python forest of [8]). Normal and unbalance windows are the main source of confusion, which the shaft-order features of the present paper resolve. The results of the present paper are obtained under the protocols of Section IV.

TABLE S1
Accuracy (%) of the Configuration of [8] and Its Variants Under the Present Code (Random Stratified 70/15/15 Split, Five Seeds)

Loads; N, overlap	Set	RF acc.	Softmax acc.	N→U (RF)	U→N (RF)
All; 32768, 50%	$\mathcal{F}_{16}$	91.2±1.4	85.7±1.3	35%	6%
All; 32768, 50%; original RF code, 10 trees	$\mathcal{F}_{16}$	88.5±6.0	–	38%	4%
All; 32768, 50%; original RF code, 1 tree	$\mathcal{F}_{16}$	86.9±6.7	–	35%	9%
All; 32768, 50%	S10	91.2±0.5	85.9±1.3	36%	6%
All; 8192, none	$\mathcal{F}_{16}$	89.7±0.9	85.6±0.7	34%	13%
All; 8192, 75%	$\mathcal{F}_{16}$	90.4±0.4	86.1±0.4	34%	10%
0 N·m; 8192, none	$\mathcal{F}_{16}$	93.1±1.0	92.3±1.3	12%	14%
2 N·m; 8192, none	$\mathcal{F}_{16}$	79.7±1.2	82.3±1.5	34%	54%
4 N·m; 8192, none	$\mathcal{F}_{16}$	94.1±1.6	93.5±0.8	7%	17%
0 N·m; 32768, 50%	$\mathcal{F}_{16}$	96.2±0.8	95.4±0.5	7%	8%
0 N·m; 32768, 50%	S10	97.2±0.6	94.9±0.7	5%	5%
2 N·m; 32768, 50%	$\mathcal{F}_{16}$	83.2±1.9	85.0±0.9	36%	40%
4 N·m; 32768, 50%	$\mathcal{F}_{16}$	95.2±1.2	95.8±0.8	9%	12%

N→U: share of normal windows predicted as unbalance; U→N: the reverse. S10: the ten spectral features of [8].

S2. Hyper-Parameter Selection

The hyper-parameters of both heads and of both updates are selected once per dataset from a grid fixed in advance, on validation data only: the validation blocks of the time-blocked folds and, for unseen loads, the windows of block 1 of the target recordings outside the support set; the test windows are never used. The grid covers the L_2 weight of the softmax head, $\lambda_2 \in \{0, 10^{-4}, 10^{-3}, 10^{-2}\}$; six RF shapes with the same flash budget of about 430 nodes (trees $\times$ depth $\times$ leaves: $5 \times 10 \times 44$, $7 \times 10 \times 32$, $10 \times 7 \times 22$, $10 \times 10 \times 22$, $14 \times 8 \times 16$, $20 \times 8 \times 11$); the on-chip update with $\eta \in \{0.03, 0.1, 0.3, 1\}$, $\lambda \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$, and $\{100, 200, 400, 800\}$ epochs; and the support weight of the gateway's RF, $\{5, 10, 20, 50, 100, 200\}$, in all 256 softmax and 36 RF configurations. Among the base models within 0.5 points of the best in-distribution validation macro-F1, the configuration with the highest mean validation macro-F1 on unseen loads (single-source and LOLO, $q = 1$ and 5) is selected (Table S2). Table S3 scores every configuration on the test windows: the selected one lies within 1.2 points of the best configuration in every cell, whereas the worst configurations fall up to 20 points lower, so the validation data identify the settings that matter.

S3. Cross-Severity Support by Direction

Table S4 splits the cross-severity results of Section V-C by direction for KAIST and reports the pooled CWRU values (three fault sizes, all ordered pairs). All entries use the same test windows within a row group: blocks 2–5 of the tested-severity recordings and of the normal recording. On CWRU, whose three fault sizes differ in their bearing-order signatures, support from another size stays at the level of the unadapted models, so labels of the tested size are the ones that count there.

TABLE S2
SELECTED HYPER-PARAMETERS

	KAIST ($\mathcal{F}_{19}$)	CWRU ($\mathcal{F}_{22}$)
Softmax head, λ_2	10^{-3}	0
RF, trees $\times$ depth $\times$ leaves	$20 \times 8 \times 11$	$7 \times 10 \times 32$
On-chip update, η / λ / epochs	0.1 / 10^{-4} / 400	0.03 / 10^{-2} / 100
Gateway RF, weight of a support window	100	10

TABLE S3
TEST MACRO-F1 (%) ON UNSEEN LOADS OF THE SELECTED CONFIGURATION AND OF THE BEST AND WORST CONFIGURATIONS OF THE GRID

Dataset	Setting	Head	$q = 1$: selected / best / worst	$q = 5$: selected / best / worst
KAIST	single-source	Softmax	89.8 / 89.8 / 73.3	96.7 / 97.2 / 77.0
	LOLO	Softmax	93.3 / 93.3 / 81.3	97.6 / 97.9 / 87.6
	single-source	RF	86.6 / 87.2 / 79.3	97.7 / 97.7 / 92.0
	LOLO	RF	95.0 / 95.0 / 87.1	98.6 / 98.7 / 95.6
CWRU	single-source	Softmax	83.6 / 84.0 / 70.9	88.8 / 90.0 / 82.2
	LOLO	Softmax	88.8 / 88.8 / 71.5	92.5 / 92.9 / 83.2
	single-source	RF	88.5 / 89.2 / 87.5	90.9 / 91.8 / 89.9
	LOLO	RF	93.9 / 94.1 / 92.4	95.3 / 95.4 / 93.2

Support from block 1, test on blocks 2–5; mean over load pairs, five folds, and three support draws. Best and worst: over the 256 (softmax) or 36 (RF) configurations of the grid, read on the test windows.

TABLE S4
MACRO-F1 (%) WITH SUPPORT FROM THE TESTED RECORDINGS OR FROM ANOTHER FAULT SEVERITY ($q = 5$; RF / SOFTMAX)

Dataset	Setting	Support $\rightarrow$ test	No adaptation	Same recordings (block 1)	Other severity (block 1)
KAIST	single-source	mild $\rightarrow$ severe	63.4 / 55.2	98.6 / 99.5	97.6 / 95.3
	single-source	severe $\rightarrow$ mild	53.9 / 49.9	98.1 / 94.4	81.4 / 79.1
	LOLO	mild $\rightarrow$ severe	92.1 / 78.4	99.5 / 100.0	98.8 / 95.8
	LOLO	severe $\rightarrow$ mild	82.8 / 77.1	98.0 / 95.2	93.6 / 88.1
CWRU	single-source	pooled	84.7 / 79.4	93.2 / 90.4	83.4 / 76.3
	LOLO	pooled	91.0 / 84.7	95.2 / 94.6	90.7 / 79.2

S4. FURTHER RESULTS

Unseen fault severities. When each fault class is trained on one severity and tested on the other at the same loads (normal split by time blocks), $\mathcal{F}_{19}$ reaches 57$\pm$10% (RF) and 50$\pm$24% (softmax) on KAIST, and $\mathcal{F}_{22}$ 46% and 42% on CWRU; the self-check gate flags these conditions with AUROC 96.9% (KAIST) and 93.5% (CWRU), so that labels for the new severity can be requested.

Additive noise and voting. White noise at 20 dB SNR lowers the clean-trained heads to 81% (RF) and 45% (softmax); noise-augmented training (30/20/10 dB copies) keeps them at 90% and 89% at 20 dB and 84% and 80% at 10 dB at no inference cost, trading clean-data macro-F1 (98.0% and 94.5%) for this robustness. Majority voting over five windows (1.6 s) raises the clean RF from 99.8% to 100% and the noise-augmented RF from 98.0% to 98.7%.

CNN inputs of 2048 samples with voting. Trained on the log-magnitude spectra of 2048-sample segments and majority-voted over the four segments of each 8192-sample window, the CNNs reach 99.5% (9.2 M parameters), 94.0% (EdgeLite), and 83.1% (MobileLite) under time-blocked evaluation and 55–68% on unseen loads, with 393, 104, and 56 KiB of activation memory.

Gate design choices. Over $W \in \{5, 10, 20\}$ on KAIST the gate AUROC for unseen loads stays at 99.0–99.4% (single-source) and 96.1–97.3% (LOLO); the per-window false-alarm rate follows the 99th-percentile threshold (2–4%). With four instead of eight prototypes per class, the AUROC is 99.0%, 94.6%, and 96.4% (single-source, LOLO, unseen severity), so eight prototypes (4.4 KiB with the inverse covariance at $D = 19$) are the operating point. Re-referencing after adaptation lowers the alarm rate on the later target blocks from 41% to 4% (single-source) and from 16% to 3% (LOLO) on CWRU, and from 91% to 4% and from 72% to 1% on KAIST.

S5. SELF-CHECK GATE ON CWRU LOADS

On CWRU, a load step from 0 to 3 hp changes the spectrum far less than on KAIST (unadapted single-source RF 88%, against 60% on KAIST), and every detector responds in proportion: AUROC 86.2% (single-source) and 70.5% (LOLO) for the KESTREL gate, against 78.7% and 63.3% for T^2 alone, 64.0% and 55.2% for disagreement plus entropy, 55.6% and 48.9%

for the maximum-probability baseline, and 84.1% and 72.2% for a nearest-neighbor search over all training windows (trailing window $W = 3$). Table S6 lists every detector on these shifts.

S6. Bearing Orders on CWRU

The bearing-order stage follows Section III-B: a fourth-order Butterworth band-pass (two biquads), the squared signal averaged over 16 samples (envelope rate 3 kHz, 512 points), a Hann window, and a real FFT. The defect orders of the SKF 6205 drive-end bearing (outer race 3.5848, rolling element 4.7135, that is, twice the ball spin, inner race 5.4152) are searched over the speed range 1730–1797 r/min at the first and second harmonics. The band is selected among four octave bands by the macro-F1 on the validation blocks (mean of both heads). Table S5 gives the time-blocked test results; every band improves on $\mathcal{F}_{19}$, and $\mathcal{F}_{16}$ with the bearing orders gives nearly the same scores as $\mathcal{F}_{22}$, so the gain comes from the bearing orders. Per-class recall rises from 96.0% to 98.8% (RF) and from 89.0% to 92.7% (softmax) for ball faults and from 83.7% to 87.9% for outer-race faults (softmax); normal windows stay at 100%. On the ESP32, the C implementation of the stage takes 1.51 ms per window (median over all 2027 CWRU windows, at most 1.60 ms) with a 2-KiB buffer and the FFT tables of the main spectrum, and the bearing orders computed on the board match the Python reference to within 1.5×10^{-6}. The ESP32-calibrated resource model, which counts 0.18×10^6 operations, predicts 3.2 ms; it therefore errs on the safe side, and its estimates for the other targets (1.5 ms on the ESP32-S3, 1.9 ms on the STM32F446, 0.4 ms on the STM32H743, and 82 ms on the RP2040) are upper bounds.

TABLE S5
Time-Blocked Macro-F1 (%) on CWRU With Bearing Orders (E3)

Features	Envelope band	Validation (both heads)	RF (test)	Softmax (test)
$\mathcal{F}_{16}$	–	–	98.2±0.8	91.2±1.8
$\mathcal{F}_{19}$	–	–	98.0±1.0	92.1±1.7
E3 only	12–20 kHz	–	83.9±1.5	79.2±1.9
$\mathcal{F}_{16}$ + E3	12–20 kHz	–	98.8±0.7	94.7±1.4
$\mathcal{F}_{22}$	1.5–3 kHz	96.05	98.3±0.8	93.8±2.1
$\mathcal{F}_{22}$	3–6 kHz	95.97	98.8±0.6	93.1±2.1
$\mathcal{F}_{22}$	6–12 kHz	96.35	98.7±0.9	94.1±1.7
$\mathcal{F}_{22}$	12–20 kHz (selected)	**96.52**	98.7±0.5	94.4±1.7

E3: the three bearing orders. Mean ± standard deviation over the five time-blocked folds.

S7. Self-Check Gate Against Other Shift Detectors

Every detector of Table S6 uses the same models, windows, trailing mean, and threshold rule (99th percentile of the trailing mean on the source validation blocks). The KESTREL gate keeps $k = 8$ prototypes per class (k-means centers of the standardized training windows) and the inverse within-class covariance; kNN keeps all standardized training windows (3816 for a single-source KAIST model, 283 KiB); the class-conditional Mahalanobis distance [30] keeps the class means and the inverse within-class covariance; PCA-SPE is the squared residual outside the principal subspace that explains 95% of the variance; the isolation forest has 100 trees. The gate is the best detector of equal or smaller memory in every row and comes within 2.2 points of kNN wherever kNN is better. At the same false-alarm budget, it also raises the most alarms among the detectors of equal memory on every KAIST shift and on unseen CWRU severities.

TABLE S6
AUROC (%) / Alarm Rate on Shifted Windows (%) of All Detectors

Shift	**KESTREL**	T^2+Dis.	T^2	Dis.+Ent.	$1 - \max p$	Energy	kNN	Class M.	PCA-SPE	IForest
KAIST load, single-source	99.2 / 91	98.2 / 83	98.3 / 83	66.0 / 49	62.2 / 40	49.6 / 19	98.9 / 85	99.1 / 89	93.2 / 73	84.7 / 12
KAIST load, LOLO	96.5 / 72	86.4 / 47	84.2 / 45	61.0 / 29	55.7 / 17	51.4 / 12	98.2 / 76	89.3 / 61	72.1 / 38	76.4 / 7
KAIST unseen severity	96.9 / 79	84.1 / 54	82.3 / 53	65.0 / 47	52.5 / 41	44.7 / 28	99.1 / 84	93.4 / 73	84.1 / 40	73.8 / 25
CWRU load, single-source	86.2 / 41	82.0 / 38	78.7 / 41	64.0 / 21	55.6 / 16	52.8 / 14	84.1 / 45	82.7 / 42	76.6 / 33	63.3 / 23
CWRU load, LOLO	70.5 / 17	64.9 / 17	63.3 / 18	55.2 / 4	48.9 / 3	47.2 / 3	72.2 / 19	65.3 / 18	63.0 / 18	58.1 / 14
CWRU unseen severity	93.5 / 60	86.5 / 40	86.3 / 39	69.4 / 28	56.9 / 6	50.8 / 7	92.3 / 42	88.2 / 45	88.3 / 36	70.3 / 3
Memory, KAIST (KiB)	4.4	1.4	1.4	0	0	0	283	1.8	1.1	≈60

Trailing window $W = 10$ (KAIST) and 3 (CWRU, shorter recordings). Means over folds and load pairs. False-alarm rates on the source test blocks: at most 5% for every detector.

Design of the gate statistic. With the same memory as T^2, a class-conditional distance already raises the KAIST LOLO and unseen-severity AUROC from 86.4% and 84.1% to 90.5% and 93.5% (distance to the mean of the predicted class; experiment e23); prototypes of the predicted class with the within-class metric reach 96.5% and 96.9% at 2% false alarms, which makes them the design of choice.

S8. Few-Shot Updates Compared

Table S7 compares the on-chip update with other updates that receive the same support windows (block 1 of the target recordings, three draws) and are scored on the same windows (blocks 2–5). Updating the full softmax head is needed: updating only the biases (cf. TinyTL [36]) leaves most of the shift uncorrected, and the prototypes of the support windows alone (nearest class mean in the standardized features) discard the source decision boundaries. Last-layer fine-tuning of the CNN baselines (Adam, 200 full-batch steps, batch-normalization statistics of the source) with the same five windows per class reaches 78.7% (EdgeLite) and 79.4% (MobileLite) on single-source KAIST, from 59.4% and 48.9% before the update (three folds, six load pairs, three support draws).

TABLE S7
Macro-F1 (%) of Few-Shot Updates of the Softmax Head on Unseen Loads (q Labeled Windows per Class)

		KAIST			CWRU		
Setting	Update	None	$q=1$	$q=5$	None	$q=1$	$q=5$
Single-source	anchored (KESTREL R)	52.6	89.8	96.7	81.0	83.6	88.8
	unanchored ($\lambda=0$)		89.8	96.7		83.7	88.9
	bias only		68.9	70.8		81.5	81.8
	nearest class mean		77.3	81.8		64.2	69.1
LOLO	anchored (KESTREL R)	76.2	93.3	97.6	86.1	88.8	92.5
	unanchored ($\lambda=0$)		93.4	97.6		88.8	92.5
	bias only		81.5	83.7		86.5	86.5
	nearest class mean		76.9	81.5		64.6	69.5

All gradient updates use the step and the number of epochs of Table S2. Mean over load pairs, five folds, and three support draws; anchored update as in Table V of the paper.

Role of the anchor. At the selected step and number of epochs, the anchor changes the macro-F1 by at most 0.1 points (Table S7); its role is to bound the drift of longer updates (experiment e27, $q=5$). With 2000 epochs at a step of 1.0, the CWRU anchor ($\lambda=10^{-2}$) halves the relative change of $\mathbf{W}$, from 1.62 to 0.76 (single-source) and from 1.16 to 0.49 (LOLO), and keeps the LOLO macro-F1 at 90.3% against 87.6% without it, so the update tolerates a larger step or a longer schedule without retuning; the weaker KAIST anchor ($\lambda=10^{-4}$) lowers the drift of such an update by 7–8%.

Labels a plant can supply. Table S8 extends Table VI of the paper with the label-free shift of healthy-baseline recalibration (HBR) and a variant that retunes only when the node's own healthy windows are misread (fewer than 90% of them predicted normal by the source head). The variant keeps the CWRU heads at their unadapted level, where faults are detected without any update, but gives up part of the single-source KAIST gain, so the paper reports the plain healthy update.

TABLE S8
Macro-F1 / Fault Detection (%) of the Softmax Head Under the Labels Available at the New Load

Labels at the new load	KAIST SS	KAIST LOLO	CWRU SS	CWRU LOLO
None	52.6 / 67.7	76.2 / 87.6	81.0 / 99.9	86.1 / 100
HBR shift (16 healthy windows, no update)	64.6 / 79.4	78.0 / 89.5	76.5 / 100	84.1 / 100
Healthy only (shift + update on 16 healthy windows)	63.7 / 85.6	71.2 / 88.1	76.5 / 100	84.1 / 100
Healthy only, retune when the healthy windows are misread	58.4 / 80.9	75.0 / 89.0	80.9 / 100	86.1 / 100
Normal + half of the fault classes ($q=5$)	81.7 / 91.0	87.1 / 93.6	80.9 / 99.9	85.8 / 99.9
All classes ($q=5$)	96.7 / 93.4	97.6 / 94.8	88.8 / 99.9	92.5 / 99.8

Fault detection: balanced accuracy of normal versus any fault. SS: single-source. Mean over load pairs, five folds, and three draws (partial and all classes).

S9. Campaigns on the ESP32

The board runs the node firmware (features, both heads, gate, on-chip update and re-referencing, run-time forest refresh); the host plays the gateway over USB (921.6 kbaud). For each single-source load pair (fold 0, one support draw), the board processes the source test block (in-distribution), block 1 of the new load (shift), 25 labeled support windows ($q=5$, block 1), and, after the update, re-referencing, and the gateway's forest and τ', blocks 2–5 of the new load. Table S9 lists the board's results; a float64 replay of the whole procedure on the host (its own update, retraining, and re-referencing on the host features of the same windows) makes the same prediction with both heads and raises the same alarm in every window of every pair.

Equivalence. With the deployed fold-0 models (all loads), the board processed every non-overlapping window of both datasets: all 8431 KAIST windows and all 2027 CWRU windows. Both heads make the decisions of the Python reference in every window, the gate statistic differs by at most 1.9×10^{-5} and raises the same alarms, and 99% of the KAIST windows reproduce every feature to within 3.6×10^{-5} (relative); on the fold-0 test windows the board reaches 99.9% (RF) and 98.1%

TABLE S9
LIFECYCLE ON THE ESP32, KAIST, SINGLE-SOURCE LOAD PAIRS (FOLD 0; %)

Load pair (N·m)	AUROC	Alarms	False alarms	Delay (windows)	F1 before, RF / SM	F1 after, RF / SM	Alarms after	Board = host
0→2	97.5	85.7	8.7	10	51.4 / 41.8	99.4 / 99.9	5.8	100
0→4	99.5	98.2	8.7	10	51.4 / 46.5	99.2 / 91.0	0.0	100
2→0	99.7	91.5	1.4	10	72.3 / 69.8	100.0 / 100.0	0.2	100
2→4	98.8	93.7	1.4	10	48.4 / 66.2	99.3 / 81.8	22.2	100
4→0	98.9	94.9	8.8	10	63.7 / 43.2	99.5 / 99.1	0.0	100
4→2	95.5	86.2	8.8	10	51.4 / 66.4	99.4 / 99.9	0.5	100
Mean	98.3	91.7	6.3	10	56.5 / 55.7	99.5 / 95.3	4.8	100

AUROC and alarms: gate statistic on block 1 of the new load against the source test block; delay: median number of windows from the start of a shifted recording to its first alarm ($W = 10$ windows are needed to fill the trailing mean). F1 before: block 1 of the new load; F1 after: blocks 2–5, after the update. Board = host: share of windows with identical predictions (both heads) and alarms. 20 374 windows in total; update 125.0 ms and re-referencing 14.1 ms per pair (means).

(softmax) on KAIST and 98.3% and 92.1% on CWRU. Per KAIST window, the features take 13.45 ms (median; at most 14.08 ms), the heads 30 μs, and the gate 133 μs with the 40 source prototypes.

Analog loop. For both models, the board first processed the fold-0 test windows digitally (in-distribution reference of the gate) and then played the fold-0 test block of each of the 27 recordings through the DAC–ADC jumper (1654 complete windows; no deadline missed). The first captured window of every recording formed the support set (27 windows); after the on-chip update, the re-referencing, and the gateway's forest and τ', the recordings were played through the loop a second time and scored without their first window (1627 windows). Noise-augmented model: gate AUROC 99.97% (digital against analog windows), alarms in every recording after the 4-window fill, macro-F1 73.6% (RF) and 46.4% (softmax) before and 71.2% and 74.7% after the update; clean model: AUROC 100%, 36.3% and 25.0% before, 81.9% and 67.1% after. On the second pass, the re-referenced gate raises alarms in 17.0% (noise-augmented) and 1.8% (clean) of the windows. With two and five labeled windows per recording, a replay of the update on the board's analog features raises the noise-augmented softmax head to 88.3% and 89.5% (tool `analog_replay`), so the number of labels per recording sets the accuracy on this coarse front end. The analog features of the clean model lie far outside the source range (standardized values up to 71), where the update becomes sensitive to rounding: a relative perturbation of 10^{-12} of its support features changes 6% of its decisions, whereas the noise-augmented model keeps every decision and the board and the float64 replay agree on all its windows. The noise-augmented model therefore serves the on-chip update, and the forest of the clean model the gateway. A 23-point DC sweep calibrates the loop (ADC $= 15.23\,\text{code} - 104$), and the I2S clock is trimmed to a measured 25 600.0 Hz.

Lifecycle on CWRU. The CWRU build of the firmware computes $\mathcal{F}_{22}$, including the bearing orders, on the chip; the lifecycle follows the KAIST procedure with $q = 5$ (20 support windows) and $W = 3$, and the verification windows are also scored before the update. Table S10 lists the twelve single-source pairs (12 168 windows). CWRU recordings are short (about eleven windows per test block), so the 99th-percentile threshold rests on few validation windows: the false-alarm rate on the source test block reaches 31% for the 0-hp models and is 0–6% for the others, and the alarm rate on the new load follows the size of the shift, as in Note S5. The softmax head of the board and a float64 replay of the update on the host make the same prediction in every window. The on-chip update takes 21.5 ms (100 epochs) and the re-referencing 16.5 ms ($D = 22$).

TABLE S10
LIFECYCLE ON THE ESP32, CWRU WITH BEARING ORDERS, SINGLE-SOURCE LOAD PAIRS (FOLD 0; %)

Load pair (hp)	AUROC	Alarms	False alarms	F1 before, RF / SM	F1 after, RF / SM	Alarms after	Board = host (SM)
0→1	70.1	47.7	31.0	85.0 / 87.0	82.9 / 87.9	11.6	100.0
0→2	82.5	78.9	31.0	83.3 / 79.1	81.2 / 84.5	6.9	100.0
0→3	88.4	80.0	31.0	81.6 / 79.9	85.9 / 83.6	0.0	100.0
1→0	100.0	100.0	0.0	89.6 / 73.0	92.4 / 95.7	12.0	100.0
1→2	74.6	13.3	0.0	94.1 / 92.7	95.6 / 93.3	1.1	100.0
1→3	90.3	22.2	0.0	90.2 / 85.1	91.2 / 85.3	0.0	100.0
2→0	99.7	89.7	3.3	87.9 / 50.8	94.3 / 80.0	19.3	100.0
2→1	86.1	4.5	3.3	96.1 / 92.9	98.0 / 94.1	3.2	100.0
2→3	75.3	4.4	3.3	93.0 / 90.6	93.9 / 90.7	0.0	100.0
3→0	97.0	100.0	5.6	83.1 / 49.1	87.6 / 95.2	10.9	100.0
3→1	85.6	25.0	5.6	85.0 / 84.6	81.4 / 88.7	6.6	100.0
3→2	72.7	22.2	5.6	92.5 / 93.1	95.9 / 92.5	4.4	100.0
Mean	85.2	49.0	10.0	88.5 / 79.8	90.0 / 89.3	6.3	100.0

F1 before and after on the same windows (blocks 2–5 of the new load). Board = host: share of windows on which the softmax head of the board and a float64 replay of the update on the host make the same prediction.

S10. Feature Ablation

Fig. S1 (left) compares feature subsets under the time-blocked protocol on KAIST. The three shaft-order features alone (O3) lift the RF to 96.1%, above the six time-domain statistics (T6, 78.6%) and the ten spectral statistics (S10 in the figure, 87.7%); the softmax head draws on the spectral context (84.8% with O3, 86.7% with S10), and adding the orders to the non-redundant statistics (F16nr) brings both heads to 99.5% and above. In the permutation test (right), permuting A_{1X} costs the RF 18.1 percentage points, nearly six times as much as any other feature; in the softmax head, the spectral entropy (15.7 points), the ratio of A_{1X} to the RMS, A_{1X}^{rel} (A_{1X}/RMS in the figure, 14.8 points), and A_{1X} (14.6 points) lead.

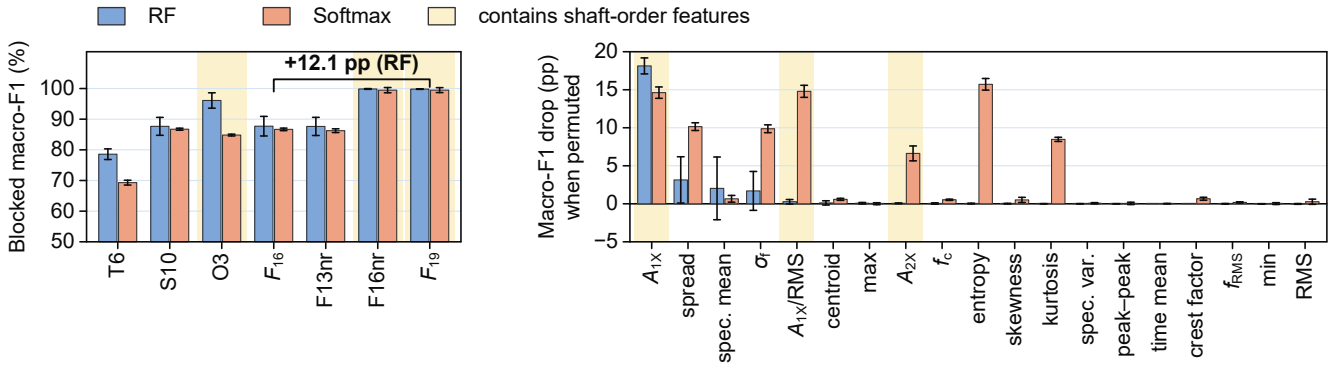


Fig. S1. Feature ablation: (left) macro-F1 of feature subsets; (right) drop of macro-F1 when one feature of $\mathcal{F}_{19}$ is permuted.